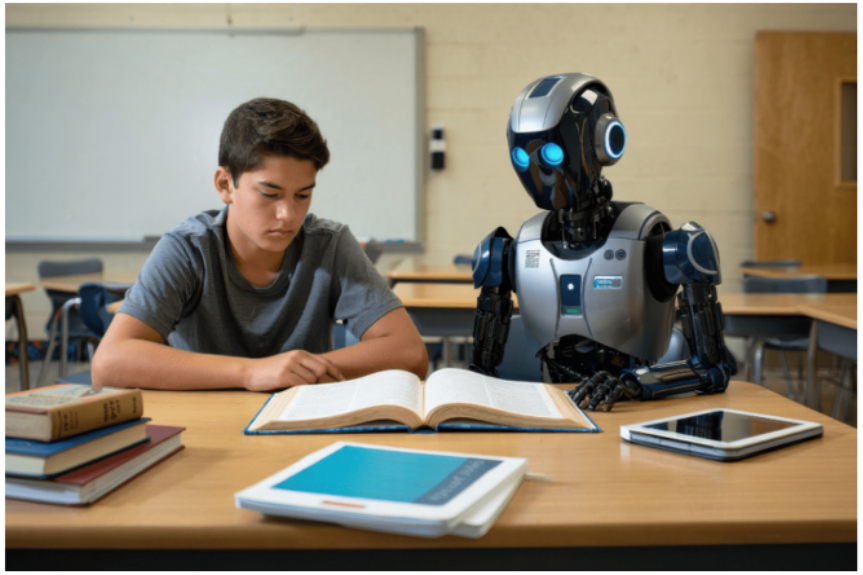
Applications of Randomized Experiment

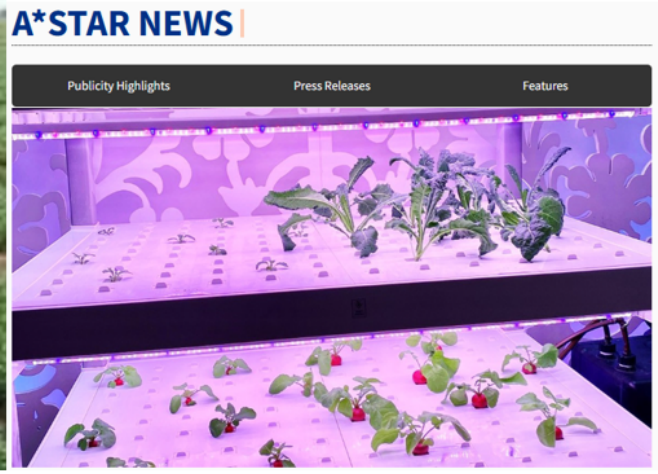


Pharmaceutical Company: New Medicine R&D

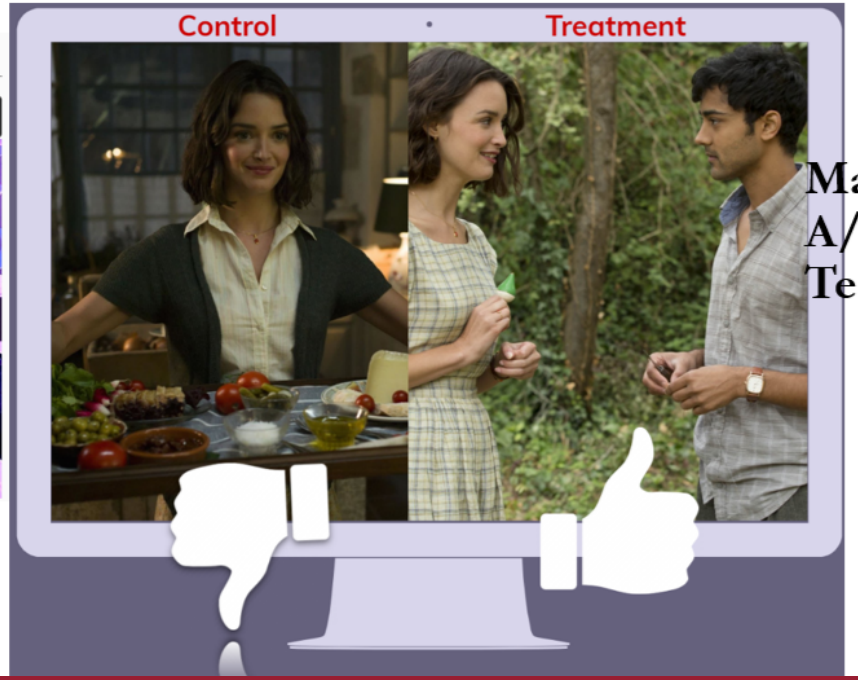
Education: AI and Learning







Agriculture: testing how different fertilizers affect crop yield



Marketing A/B Testing

Why Do We Need Randomized Experiments?

Prediction: Which customers are likely to click?
Intervention: Will changing the button make more customers click?

Correlation: a statistical relationship between random variables

Causation: Changing X leads to a change in Y

Famous definition by John Stuart Mill and is widely quoted in social science; causal relationship satisfies three criteria:
1) The "cause" precedes the "effect" in time (temporal precedence),
2) The "cause" and the "effect" are related (covariation), and
3) There are no plausible alternative explanations for the observed covariation (non-spuriousness).



mechanism

outcome → $y = f(x)$ cause

Correlation and Causal Claims

An almost perfect match

r = 0.993

More Python, more wind?

Spurious Correlations by Tyler Vigen

tylervigen.com/spurious-correlations

Correlation and Causal Claims

1,000 variables

499,500

different pairs

More comparisons, more chances to find a match.

Spurious Correlations by Tyler Vigen

tylervigen.com/spurious-correlations

Correlation (by Coincidence or Unknown Factors)

Psychology of Popular Media Culture

© 2014 American Psycho
2160-4134/14/$12.00 DOI: 10.

Violent Video Games and Real-World Violence: Rhetoric Versus Data

Patrick M. Markey
Villanova University

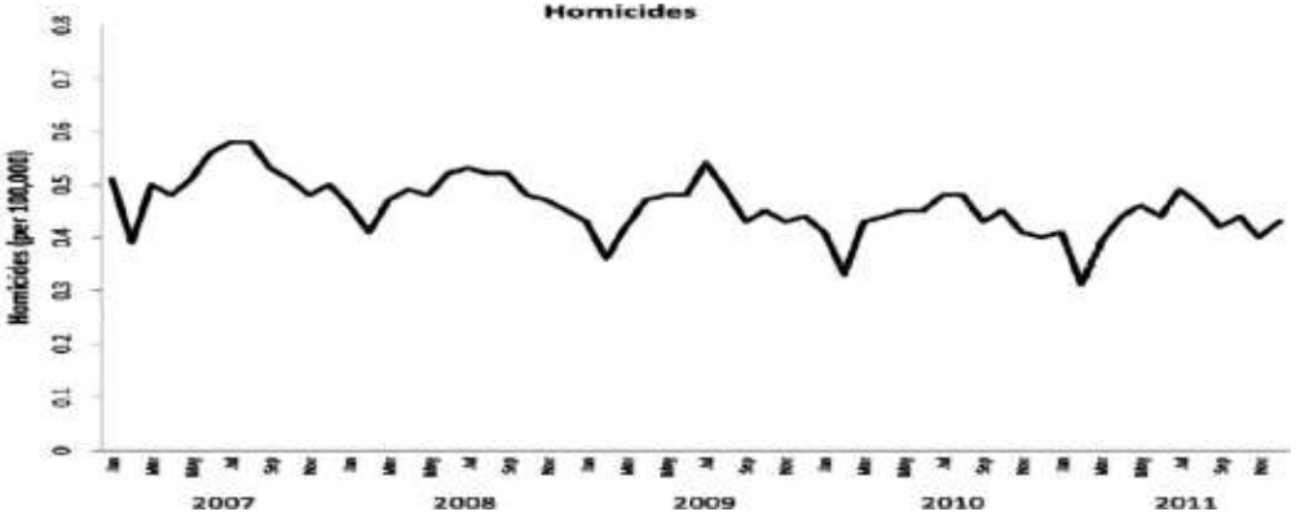
Charlotte N. Markey
Rutgers University

Juliana E. French
Villanova University

Laboratory and correlational studies often fi a link between violent video games and minor or benign forms of aggressive behaviors (e.g., exposing an opponent to an unpleasant noise). Based on these studies, the media, lawmakers, and researchers often imply a link between violent video games and violent criminal behavior. Using a similar methodology employed by researchers to examine predictors of severe violent behaviors (Anderson et al., *Journal of Personality and Social Psychology* 73: 1213–1223, 1997), 4 time-series analyses investigated the associations among violent crime (homicides and aggravated assaults), video game sales, Internet keyword searches for violent video game guides, and the release dates of popular violent video games (both annually and monthly). Contrary to the claims that violent video games are linked to aggressive assaults and homicides, no evidence was found to suggest that this medium was positively related to real-world violence in the United States. Unexpectedly, many of the results were suggestive of a *decrease* in violent crime in response to violent video games. Possible explanations for these unforeseen fi are discussed and researchers are cautioned about generalizing the results from laboratory and correlational studies to severe forms of violent behavior.

Keywords: video games, time series, violence, homicide, ARIMA





Correlation (by Coincidence or Unknown Factors)



© 2014 American Psycho
2160-4134/14/$12.00 DOI: 10

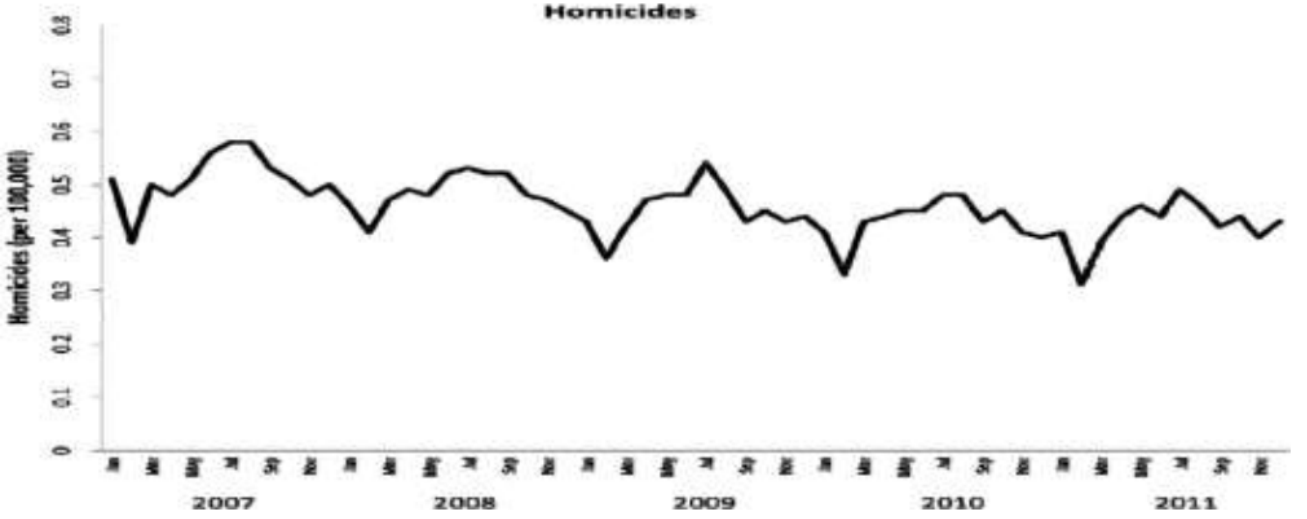
orld Violence:
a

Charlotte N. Markey
Rutgers University

en violent video games and
posing an opponent to an
awmakers, and researchers
violent criminal behavior.
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tively related to real-world
results were suggestive of
games. Possible explana-
researchers are cautioned
elational studies to severe

A





Leo Tolstoy once said in *Anna Karenina* that, "Happy families are all alike; every unhappy family is unhappy in its own way"

An anonymous statistician (who loves Russian literature) says, "All happy statisticians are happy in their own way; but all the unhappy ones are alike — they all do causal inference with observational data."

An Example

I have a new medicine that is supposed to reduce blood pressure.

I want to establish the causal relationship between between this medicine and blood pressure reduction



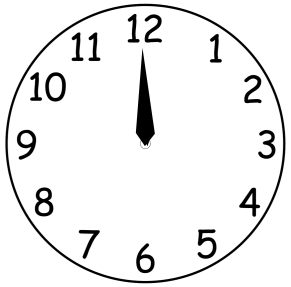
I propose the following experiment:



180









140



A patient with high blood pressure.

Systolic blood pressure (mmHg). Numbers and timing are illustrative.

An Example

There are environmental (external) factors that can confound the results.

e.g., we don't know what has happened during these 30 minutes: temperature may have changed, humidity may have changed, etc…

You can never control all the environmental factors in this experiment. Hence, you cannot claim that the reduction in this patient's blood pressure is due to the medicine.

What should we do?

An Example

We find another person and put him in the same room with our patient.

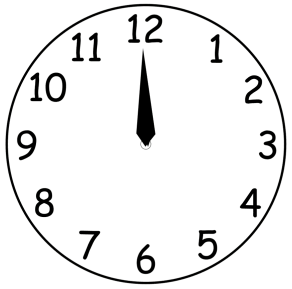


180



A patient with high blood pressure







145



Another one

140



138



An Example

The other one in this experiment is a "control"



180



A patient with high blood pressure

Difference is 40



145



Control

Difference is 7

Environmental variable may have some effect

140

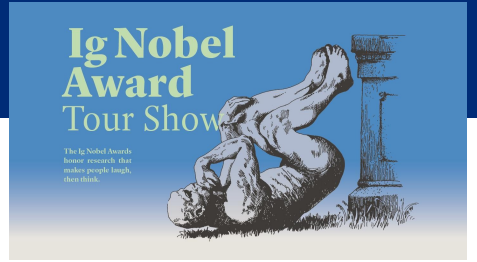


Difference in
Difference is 40-7=33

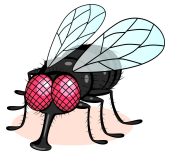
138



Choosing the Right Control



2019 study I 2025 Ig Nobel Biology Prize



No paint



Black-and-white stripes



Black paint on black fur



Could the paint itself explain fewer flies?

About 50% fewer fly landings with zebra stripes

Choose a control that checks an alternative explanation.

Photos: Kojima et al., PLOS ONE (2019), Fig. 1, CC BY 4.0. Panels c, a, b. Study: 6 cows, each in all 3 conditions.

An Example

Give the "control" a sugar pill and tell him it is the same medicine received by the patient



180





A patient with high blood pressure

"Placebo Effect"





145



Control

Difference is 40

Difference is 7

140



Difference in Difference is 40-7=33

138



An Example

Give the "control" a sugar pill and tell him it is the same medicine received by the patient



A patient with high blood pressure

180





Difference is 40

140





"Placebo Effect"



Control

145



Difference is 7

Difference in
Difference is 40-7=33

138



An Example

The "control" guy is NOT the patient

e.g., the "control" guy is 10 years older than the patient



180



A patient with high blood pressure



Difference is 40

140



e.g., the "control" guy is heavier than the patient



145





Control

Difference is 7

Difference in

Difference is 40-7=33

138



An Example

The "control" guy is NOT the patient

Patient



180





-40

140

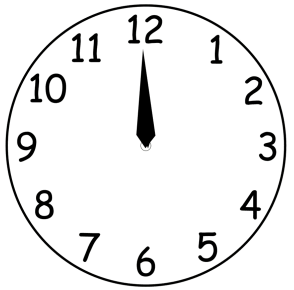


Control, 10 years older than the Patient



145







-7

138



An Example

The "control" guy is NOT the patient

Patient



180

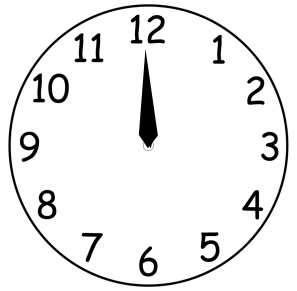
Environmental Change

185



140

-45



Control, 10 years older than the Patient



145

Environmental Change

180



138

-42

An Example

The "control" guy is NOT the patient

Patient

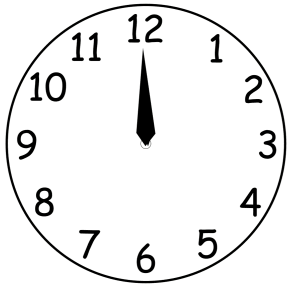


Environmental Change

180 → 185 → 140



-45



Control, 10 years older than the Patient



Environmental Change

145 → 150 → 138



-12

An Example

The "control" guy is NOT the patient

Patient



180

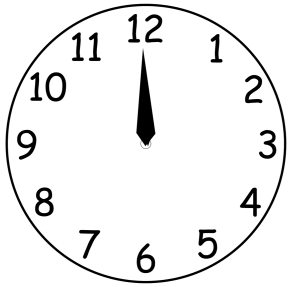
Environmental Change



185

140

-45



Control, same age as the Patient



Environmental Change

187

141

-46



An Example

Clone the patient

Patient



Environmental Change

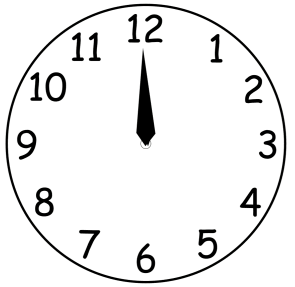
180



185

140

-45



Clone Patient



Environmental Change

180



185

182

-3

An Example

The "control" guy is NOT the patient

e.g., the "control" guy is 10 years older than the patient



A patient with high blood pressure

e.g., the "control" guy is heavier than the patient



Control

What should we do?

Potential Outcome Framework

The answer is both "Yes" and "No"

It is impossible to make causal statement about ***individual*** subject

Potential Outcome Framework

- Each level of the treatment (or manipulation) corresponds to one potential outcome of each individual subject
- Each of these potential outcomes describes what we would observe under that treatment
- We observe the outcome under the treatment actually received. The alternative remains unobserved

Potential Outcome Framework

The same subject at a different time is a different subject

If one could rewind history, and change only the action, then causation could be observed.

"No man ever steps in the same river twice, for it's not the same river and he's not the same man." (Commonly attributed to Heraclitus)

"There is sometimes a tendency to view the same physical object at different times as the same unit. ***We view this as a fundamental mistake****. The same physical unit, "myself at different times," is not the same unit in our approach to causality. Time matters for many reasons. … It is often reasonable to assume that time makes little difference for inanimate objects –we may feel confident, from past experience, that turning on a faucet will cause water to flow from that tap – but this assumption is typically less reasonable with human subjects, and it is never correct to confuse assumptions (e.g., about similarities between different units), with definitions (e.g., of a unit, or of a causal effect)."----------Rubin*

Definition of Causal Effects

Unit	Potential Outcomes		Causal Effect
	$Y(A)$	$Y(B)$	
You	37	5	Improvement due to Treatment A (37-5=32)

A and B are 2 medicines for reducing blood pressure

Y is the reduction in blood pressure

Definition of Causal Effects

1. The definition of the causal effect depends on the potential outcomes, but it *does not* depend on which outcome is actually observed.

2. The causal effect is the comparison of potential outcomes for the *same unit*, at the *same moment* in time post-treatment.

Definition (Treatment)

D_i: Indicator of treatment intake for unit (or subject) i, where $i = 1, ..., N$

$$D_i = \begin{cases} 1 & \text{if unit } i \text{ received the treatment,} \\ 0, & \text{otherwise} \end{cases}$$

Definition of Causal Effects

Definition (Observed Outcome)

Y_i^{obs} : The observed outcome of subject i

Definition (Potential Outcome)

$Y_i(1)$: Potential outcome for unit i with treatment
$Y_i(0)$: Potential outcome for unit i without treatment

Observed outcomes are realized from potential outcomes as

$$Y_i^{obs} = D_i Y_i(1) + (1 - D_i) Y_i(0) = \begin{cases} Y_i(1), & \text{if } D_i = 1, \\ Y_i(0), & \text{if } D_i = 0. \end{cases}$$

Definition (Causal Effect, or Unit Treatment Effect)

The causal effect of the treatment on the outcome for unit *i* is the difference between its two potential outcomes:

$$\tau_i \triangleq Y_i(1) - Y_i(0)$$

Fundamental Problem of Causal Inference (Holland 1986): "We can never observe both $Y_i(1)$ and $Y_i(0)$ for the same *i*"

"The problem of causal inference is a missing data problem: given any treatment assigned to an individual unit, the potential outcome associated with any alternate treatment is missing." (Rubin 1974)

It is impossible to identify τ_i without making further assumptions.

Definition of Causal Effects

"Solutions": assuming $Y_i(1)$ are constant across all individuals, and and $Y_i(0)$ are constant across all individuals, then cross-sectional comparisons will recover $\tau_i = Y_i(1) - Y_j(0)$

"Solutions": assuming $Y_i(1)$ are constant across time, and $Y_i(0)$ are constant across time, then before-and-after comparisons will recover τ_i

These might be sometimes plausible in physics sciences

Unfortunately, rarely true in social sciences

Unit-level causal effects are fundamentally unobservable. We instead focus on population average causal effect

Recall: $Y_i^{obs} = D_i Y_i(1) + (1 - D_i) Y_i(0)$

This notation implicitly makes the following assumption:

Stable Unit Treatment Value Assumption (SUTVA)

1. No interference between units

➢Potential outcomes for a unit must not be affected by the treatment of any other units

❑ *Violations: spill-over effects, contagion*

2. No different versions of treatment

➢Nominally identical treatments are in fact identical

❑ *Violations: variable levels of treatment*

Causal Inference without SUTVA

Assume $i = 1, 2$ and $D_i \in \{0, 1\}$

What are the potential outcomes for subject 1 under SUTVA? $Y_1(0), Y_1(1)$

What are the possible values of the vector (D_1, D_2)?

$$(D_1, D_2) \in \{(0,0), (1,0), (0,1), (1,1)\}$$

What are the potential outcomes for subject 1?

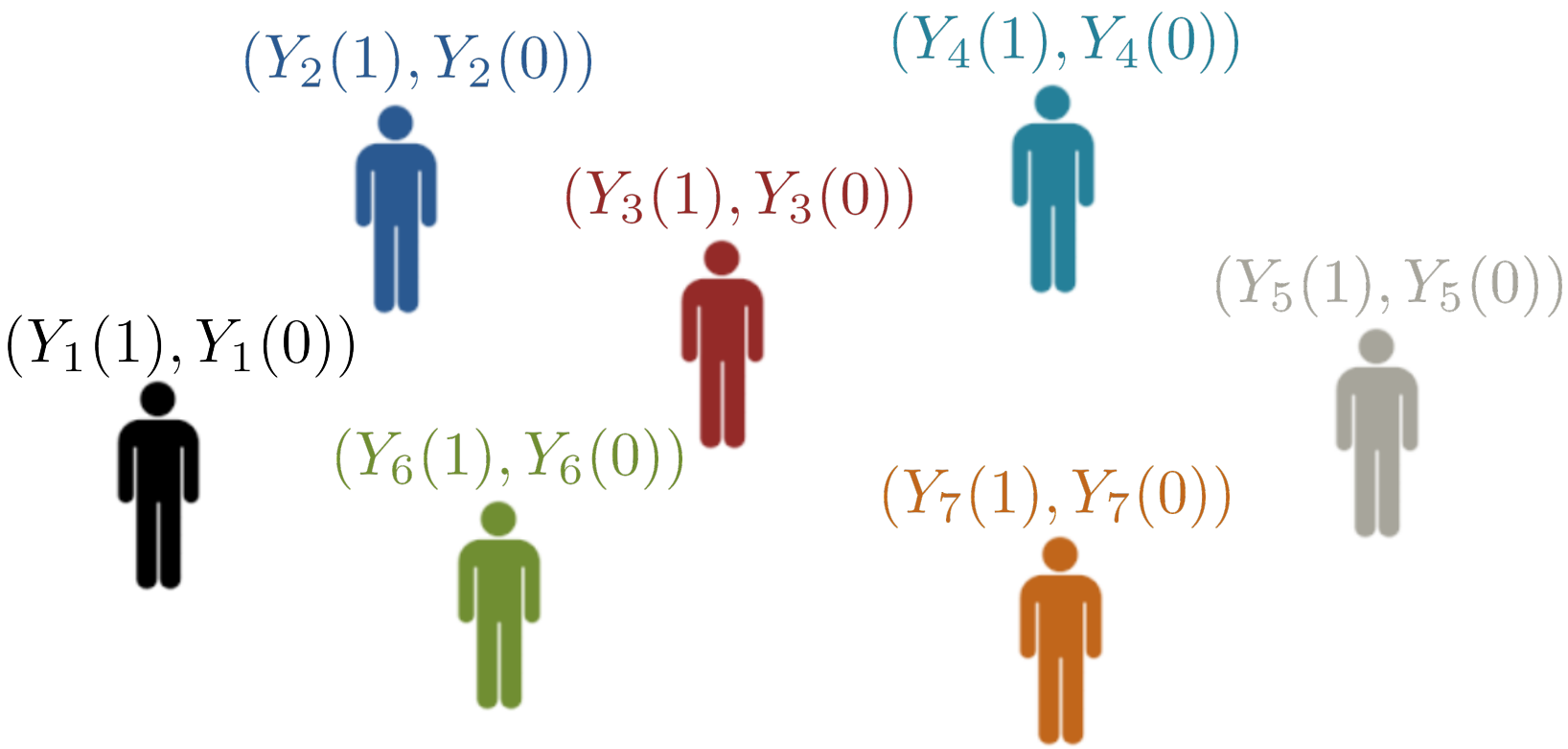
$$Y_1((0,0)),\ Y_1((1,0)),\ Y_1((0,1)),\ Y_1((1,1))$$

What are the causal effects for subject 1?

$$Y_1((1,1)) - Y_1((0,0)),\ Y_1((1,1)) - Y_1((0,1)),\ Y_1((1,1)) - Y_1((1,0))$$
$$Y_1((1,0)) - Y_1((0,0)),\ Y_1((1,0)) - Y_1((0,1)),\ Y_1((0,1)) - Y_1((0,0))$$

Without SUTVA, causal inference becomes exponentially more difficult as N increases. For our purpose, we always assume SUTVA holds.

Population Average Causal Effect



Assume there are N people in the population, then the ***average causal effect*** of the population is

$$\tau_{fs} = \frac{1}{N}\sum_{i=1}^{N}[Y_i(1) - Y_i(0)] = \frac{1}{N}\sum_{i=1}^{N} Y_i(1) - \frac{1}{N}\sum_{i=1}^{N} Y_i(0) = \bar{Y}(1) - \bar{Y}(0)$$

Estimation of Population Average Causal Effect

In definition of causal effect, we need to compare the ***potential*** outcomes

But you are only able to observe at most one potential outcome for each individual

In estimation of causal effect, we can only use the ***observed*** outcomes

Unit	Potential Outcome		Causal Effect
	$Y_i(1)$	$Y_i(0)$	$Y_i(1)-Y_i(0)$
Subject #1	4	22	-18
Subject #2	20	13	7
Subject #3	30	19	11
Subject #4	11	22	-11
Average	16.25	19	-2.75

Does AI Access Improve Debugging Performance?

Four students complete the same debugging exercise under the same time limit.
Treatment 1: access to an AI coding assistant.
Treatment 0: standard coding tools without the AI assistant.
Outcome: debugging score, out of 30 points.

What is the average causal effect? $\tau_{fs} = \bar{Y}(1) - \bar{Y}(0)$

Estimation of Population Average Causal Effect

Unit	Potential Outcome	
	$Y_i(1)$	$Y_i(0)$
Subject #1		22
Subject #2	20	
Subject #3	30	
Subject #4		22

Treatment	***Observed Outcome***
0	22
1	20
1	30
0	22

1 = AI access; 0 = no AI access. Outcome: debugging score / 30.

How should we estimate the population average causal effect?

$$\tau_{fs} = \bar{Y}(1) - \bar{Y}(0)$$

$$\hat{\tau}_{fs} = \bar{Y}_T^{obs} - \bar{Y}_C^{obs}$$

Estimation of Population Average Causal Effect

Unit	Potential Outcome	
	$Y_i(1)$	$Y_i(0)$
Subject #1	[redacted]	22
Subject #2	20	[redacted]
Subject #3	30	[redacted]
Subject #4	[redacted]	22

Treatment	***Observed Outcome***
0	22
1	20
1	30
0	22

1 = AI access; 0 = no AI access. Outcome: debugging score / 30.

What is the difference in average observed outcomes between treatment 1 and treatment 0?

Estimation of Population Average Causal Effect

Unit	Potential Outcome	
	$Y_i(1)$	$Y_i(0)$
Subject #1	4	
Subject #2		13
Subject #3		19
Subject #4	11	

Treatment	***Observed Outcome***
1	4
0	13
0	19
1	11

1 = AI access; 0 = no AI access. Outcome: debugging score / 30.

What is the difference in average observed outcomes between treatment 1 and treatment 0?

Estimation of Population Average Causal Effect

Unit	Potential Outcome	
	$Y_i(1)$	$Y_i(0)$
Subject #1	4	
Subject #2		13
Subject #3		19
Subject #4	11	

Treatment	*Observed Outcome*
1	4
0	13
0	19
1	11

Unit	Potential Outcome	
	$Y_i(1)$	$Y_i(0)$
Subject #1		22
Subject #2	20	
Subject #3	30	
Subject #4		22
Average	16.25	19

Treatment	*Observed Outcome*
0	22
1	20
1	30
0	22

Assignment Mechanism

Why does this happen?

Because the assignment mechanism matters

In both of these estimates, ***the assignment mechanism is possibly related to the potential outcomes.***

- In the 1st estimate, each subject was given the most effective treatment
- In the 2nd estimate, each subject was given the least effective treatment

Using observed outcomes to do causal inference is cherry-picking if the assignment mechanism is not well chosen.

Balance Two Groups

For each subject, you flip an unbiased coin. If it is head, then the subject goes to one group; otherwise, they go to another group.





Treatment 1: AI Access

Treatment 0: No AI Access

This ensures that ex ante, each subject has an equal chance of being in each group, and each subject's assignment is independent of other subjects. More importantly, each subject's assignment is independent of their potential outcome

Flipping a Biased Coin



Treatment 1:
AI Access

What if I flip a biased coin, with a 45% chance of landing head?

What if I flip a biased coin, with a 10% chance of landing head?

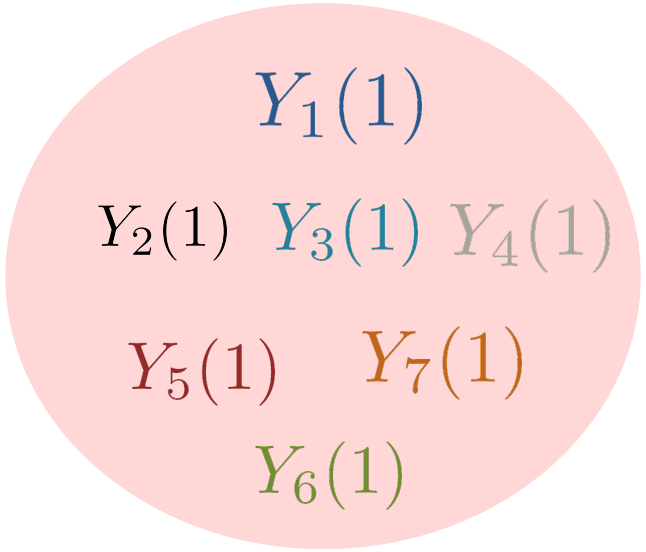
Treatment 0:
No AI Access

Representatives of Each Population

$$\tau_{fs} = \bar{Y}(1) - \bar{Y}(0)$$

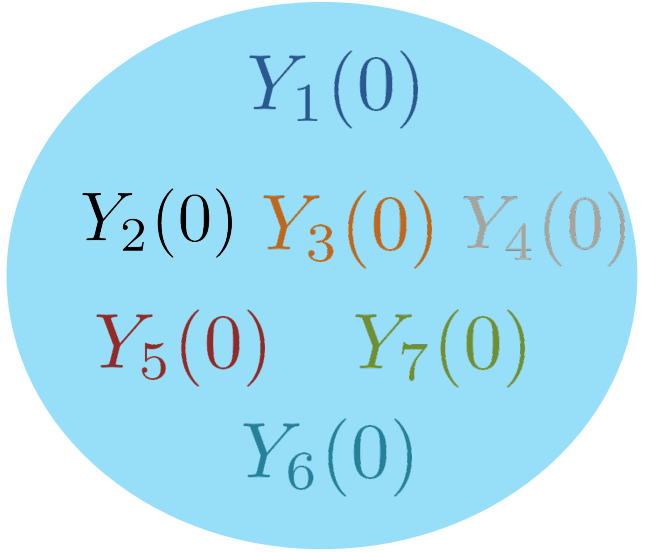
Thus, τ_{fs} can be viewed as the difference between a "representative" subject under treatment 1 and a "representative" subject under treatment 0

Treatment 1



Except for taking the average, are there other ways to get a "representative" subject from each distribution?

Treatment 0



If I just blindly draw some sample(s) from 1, e.g., $Y_j(1)$ and $Y_i(1)$, the corresponding sample average is a representative subject of distribution *Y1*

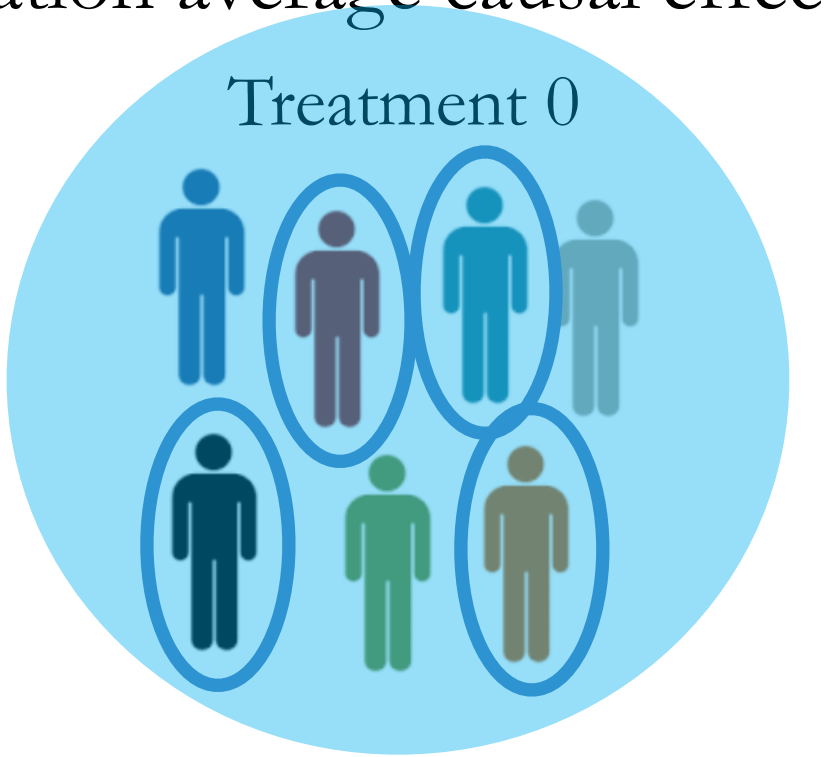
Same idea for Y0, randomly draw $Y_k(0)$, $Y_m(0)$

Representatives of Each Population

In our setting, the goal is to derive the population average causal effect



In each population, you only got a sample



$$\hat{\tau}_{fs} = \bar{Y}_T^{obs} - \bar{Y}_C^{obs}$$

$\approx$ ***Hopefully***

$$\tau_{fs} = \bar{Y}(1) - \bar{Y}(0)$$

$$\bar{Y}_T^{obs} = \frac{1}{N_T} \sum_{i \in T} Y_i(1)$$

$$\bar{Y}_C^{obs} = \frac{1}{N_C} \sum_{i \in C} Y_i(0)$$

The condition for "$\approx$" is: the assignment is independent of the potential outcome